HW4
CFD, IISC
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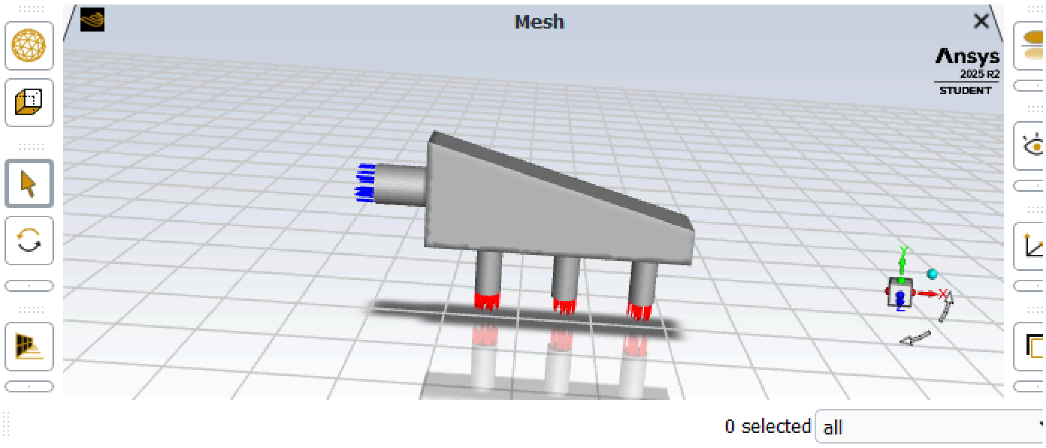
A screen capture of the mesh quality information



Console

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Domain Extents:
  x-coordinate: min (m) = -1.270000e-02, max (m) = 6.350000e-02
  y-coordinate: min (m) = -1.270000e-02, max (m) = 2.527300e-02
  z-coordinate: min (m) = 0.000000e+00, max (m) = 1.270000e-02
Volume statistics:
  minimum volume (m3): 4.678409e-12
  maximum volume (m3): 9.008680e-10
    total volume (m3): 1.560530e-05
Face area statistics:
  minimum face area (m2): 1.459336e-08
  maximum face area (m2): 2.112229e-06
Checking mesh............................
Done.

Mesh Quality:

Minimum Orthogonal Quality =  1.17613e-01 cell 1961 on zone 3 (ID: 4599 on partition:
0) at location ( 9.43755e-06,  1.50988e-02,  1.01999e-02)


Maximum Aspect Ratio =  1.24789e+02 cell 1930 on zone 3 (ID: 4581 on partition: 0) at
location ( 8.32724e-06,  1.04644e-02,  8.50250e-04)
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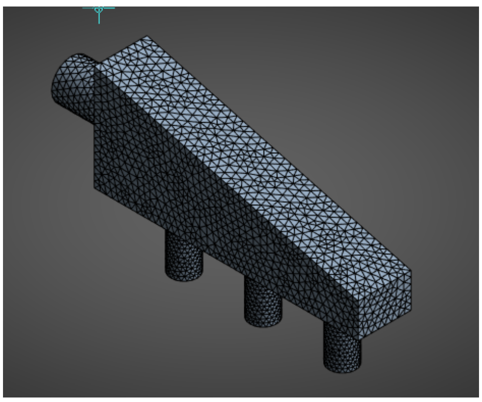
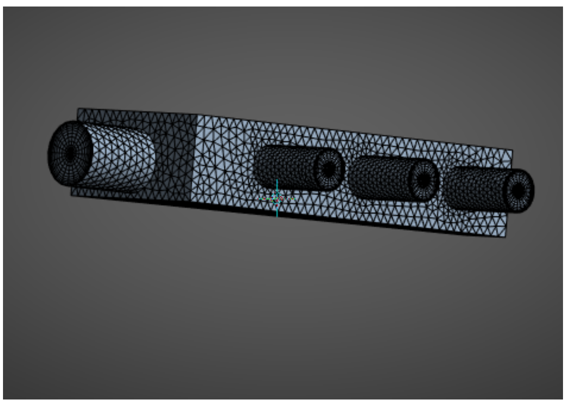



- The corresponding average inlet velocity in m/s which matches a flow rate of 3 l/min.

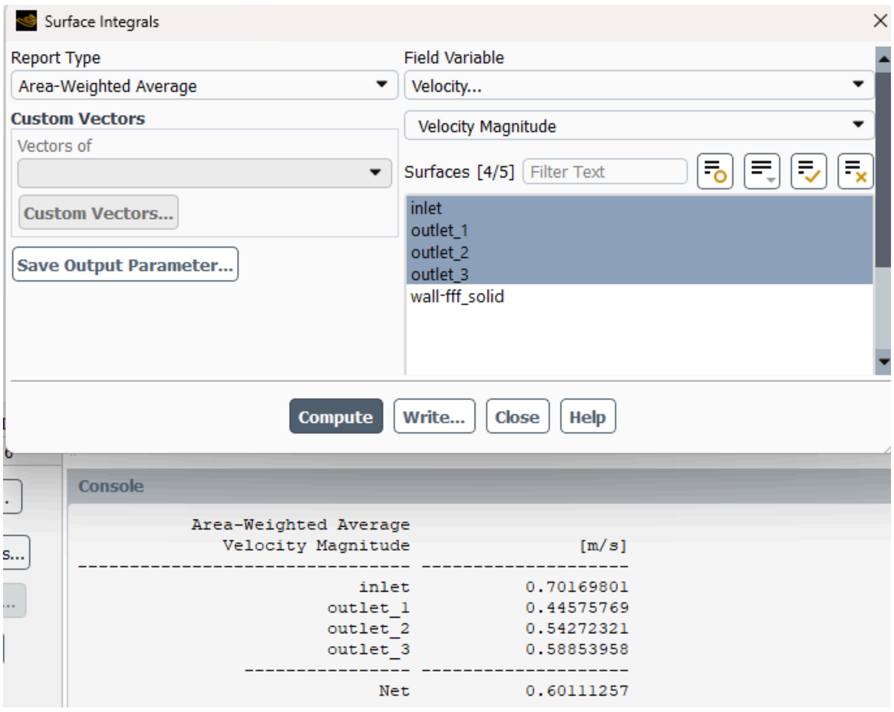
$Q = 3$ L/min $= 3 \times 10^{-3}$ m³/min $= (3 \times 10^{-3}) / 60 = 5 \times 10^{-5}$ m³/s

$D = 0.375$ inch $= 0.375 \times 0.0254 = 0.009525$ m

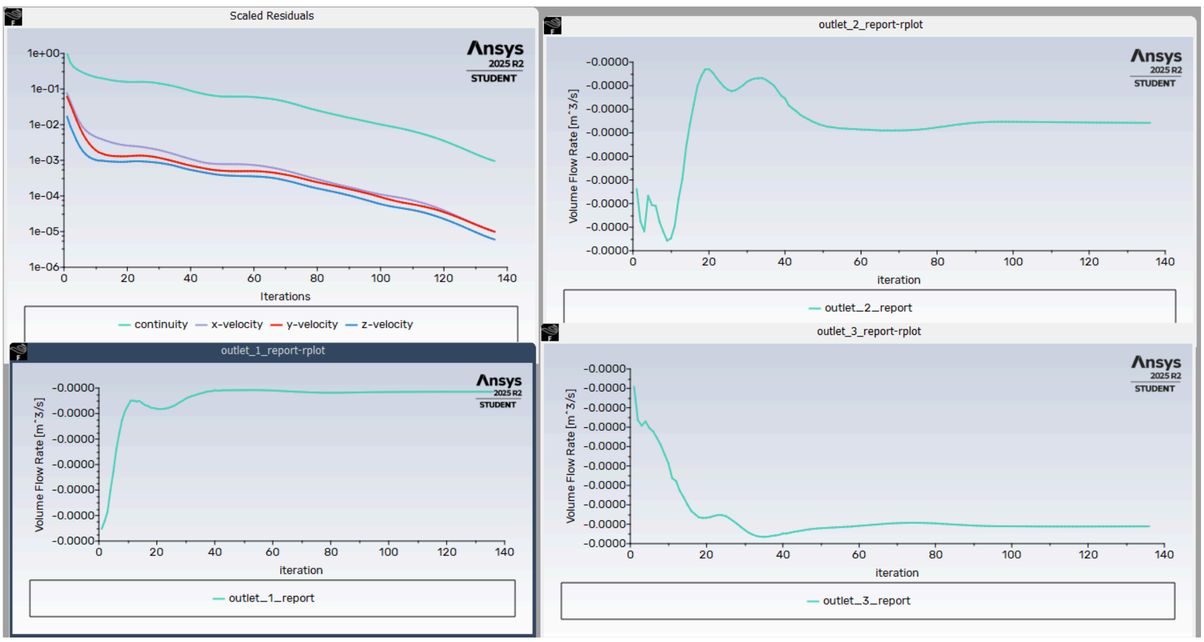
$A = (\pi \times D^2) / 4 = (\pi \times 0.009525^2) / 4 = 7.12 \times 10^{-5}$ m²

$V = Q / A = (5 \times 10^{-5}) / (7.12 \times 10^{-5}) \sim 0.70$ m/s

Average inlet velocity = 0.70 m/s

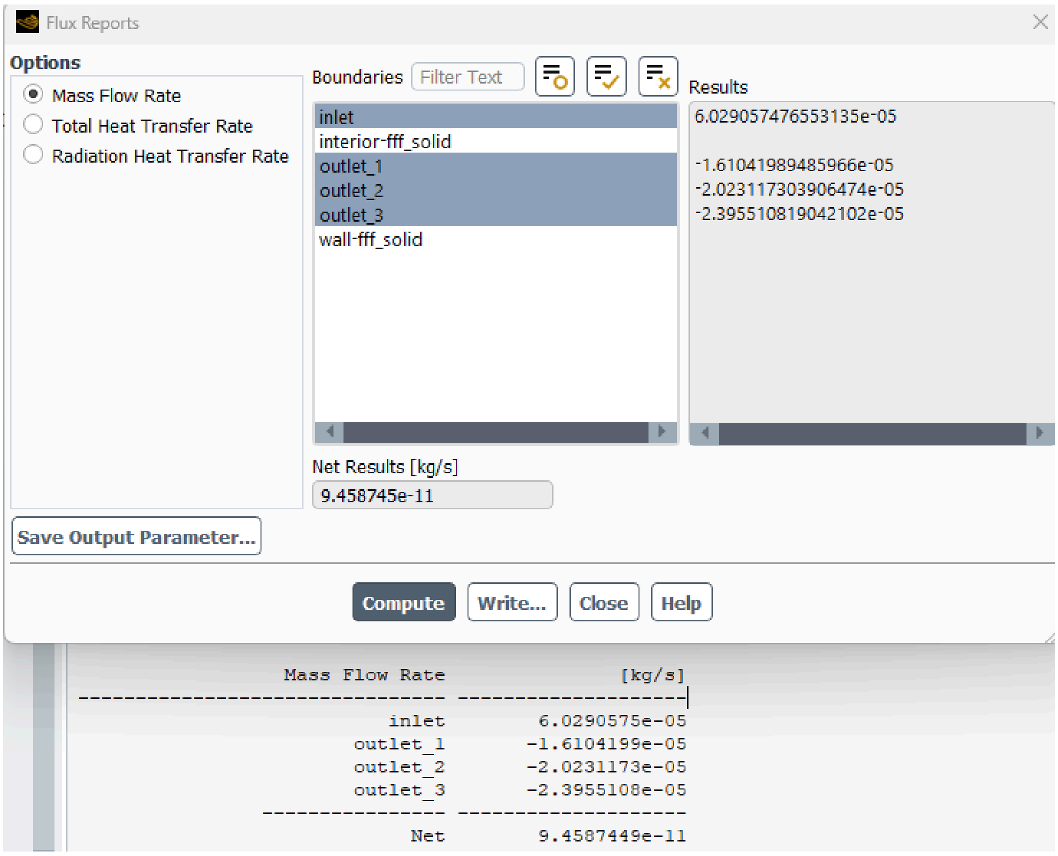


A screen capture of the decreasing residuals the flow rate through the 3 outlets



- A screen capture showing that the inlet and outlet mass fluxes match within lessthan 1%

Error ≈ $(9.46 \times 10^{-11} / 6.03 \times 10^{-5}) \times 100 \approx 0.00000016\%$



A screen capture of the particle pathlines.